

Trust the messenger, then the message: unpacking the cross-route logic of danmaku engagement

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Abstract

Purpose – Contemporary video repositories increasingly overlay real-time, user-generated metadata in the form of danmaku (bullet comments) onto linear content, creating a complex retrieval environment defined by the tension between collaborative sensemaking and cognitive overload. This study examines this human information interaction dynamic, analyzing how observable creator-related source cues, Content value (CV) signals and peer feedback are associated with engagement within information-dense streams.

Design/methodology/approach – Leveraging large-scale behavioral trace data from 261 informational videos and over 130,000 time-anchored comments on Bilibili, the study employs a multi-method approach combining partial least squares structural equation modeling, partial least squares multi-group analysis and BERTopic modeling to triangulate structural engagement associations with semantic discourse patterns.

Findings – The analysis identifies association patterns consistent with a sequential cross-route account in which peripheral source cues function as heuristic filters that are linked to CV signals and subsequent feedback interaction. Crucially, this pattern is context-contingent: high-efficacy information environments show stronger content-oriented engagement, whereas low-efficacy contexts show stronger associations with peripheral source heuristics and social signals.

Originality/value – The research refines dual-process theories within digital information management by specifying a measurable, sequential association pattern for observable engagement under information density. It distinguishes Audience Conversion Efficacy as a critical moderator, offering evidence-based strategies for adaptive interface design that align information presentation with content utility.

Keywords Elaboration likelihood model, Human information interaction, Behavioral trace data, User engagement, Social video platforms

Paper type Research article

1. Introduction

Video-sharing platforms increasingly function as information environments in which users evaluate content, interpret platform-visible cues and respond to peer reactions within the same viewing interface (Amaldoss and Shin, 2026). Bilibili provides a relevant setting for examining this process because its danmaku (bullet comments) system overlays time-synchronized viewer messages directly onto the video stream (Li and Tao, 2026). This feature turns video viewing into a layered information process in which the primary video content is accompanied by a continuous stream of social annotation. At the behavioral level, danmaku



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comment density on Bilibili is associated with users' viewing and awarding behavior and this association may become non-linear above a density threshold (Li *et al.*, 2025). At the socio-cognitive level, viewer cues and danmaku timing are linked to social cognition and narrative engagement on online video platforms (Zhang *et al.*, 2025b). Taken together, this evidence positions danmaku not merely as an interaction feature but as part of the information environment through which users evaluate whether and how to engage.

This information layer is especially consequential in educational video contexts, where engagement should support effortful information processing rather than only social co-presence. Danmaku can enhance perceived togetherness and prompt peer exchange (Xiang and Chae, 2022), functioning as a social media affordance that facilitates knowledge sharing within the learning community (Li *et al.*, 2024a), but it can also introduce competing information that fragments attention and intensifies cognitive overload (Peng *et al.*, 2025). In instructional video learning, danmaku interruptions can distract learners and increase cognitive load for some cognitive styles (Mu *et al.*, 2025). This tension is central to human information interaction (HII) because viewers encounter competing creator-related source cues, content value (CV) signals and peer feedback while deciding whether to remain passive viewers or become real-time contributors. Although prior work offers valuable insights, two gaps remain. First, existing studies often examine these cues in isolation or as parallel predictors, which provides limited explanation of how users combine multiple signals when navigating high-density environments (Dai *et al.*, 2026). Second, research based mainly on self-reported perceptions may diverge from observable behavior in naturalistic platform settings (Costa-Feito *et al.*, 2025). Together, these limitations motivate an association-based account of danmaku engagement as a sequential HII process in which users' source screening, content evaluation, peer feedback and real-time participation are modeled as linked components of engagement.

We ground the model in the elaboration likelihood model (ELM) (Petty and Cacioppo, 1986) and use the community of inquiry (COI) framework (Garrison *et al.*, 2001) as a contextual lens to interpret learning-oriented participation. Methodologically, we analyze a large-scale archival dataset of 261 educational videos and integrate partial least squares structural equation modeling (PLS-SEM), multi-group analysis (MGA) and BERTopic modeling of more than 130,000 real-time comments to triangulate structural relations with discourse patterns.

This study therefore addresses the following central research questions: (1) How do peripheral cues (source credibility (SC)) and central cues (CV and peer feedback) interact to shape user engagement in user-generated content (UGC) video environments, and can a sequential, cross-route model better explain this dynamic than traditional parallel-processing frameworks? (2) Under what specific conditions does this engagement pattern emerge, and how is the intrinsic efficacy of the video content associated with differences in these structural paths? (3) What underlying behavioral signature, as revealed through users' real-time discourse, is associated with this moderation?

In answering these questions, this paper makes three contributions to information management and HII research. First, it specifies and tests a sequential cross-route association pattern showing how source-related cues, content evaluation signals, and peer feedback jointly relate to real-time engagement (RTE) in social video environments. Second, it specifies boundary conditions by showing that key path strengths differ across content contexts with high versus low audience conversion efficacy (CE), clarifying when observable engagement is more closely aligned with source cues versus content-centric interaction. Third, it triangulates structural results with discourse evidence by linking moderation patterns to thematic differences in danmaku, offering an ecologically valid account of how engagement patterns vary under information density.

2. Theoretical background and hypotheses

2.1 *Danmaku as a dynamic information layer*

Danmaku refers to time-anchored viewer messages that overlay the video screen, functioning as a distinct secondary information layer during content consumption (Li, 2017). Unlike traditional asynchronous comments, this feature externalizes viewers' in-the-moment interpretations and exposes them to continuous peer-produced cues that can redirect attention and reshape ongoing information processing (Yuan *et al.*, 2025). This dual character makes danmaku a useful behavioral trace for information management research, as it records time-stamped moments of active contribution under conditions of high information density. Consequently, rather than viewing engagement solely as social participation, we conceptualize it as a sequence of information filtering and processing decisions in which creator-related source cues, CV signals and peer feedback are jointly present.

2.2 *Elaboration likelihood model and conceptual framework*

To explain engagement in this cue-rich environment, we situate the ELM within the broader tradition of dual-process theories of information processing. Dual-process accounts distinguish relatively fast, heuristic processing from more effortful, analytic processing (Evans, 2008). The ELM applies this distinction to persuasion and information evaluation by proposing that individuals rely on a central route when they scrutinize message or content arguments, and on a peripheral route when they use cues such as SC, popularity, or social endorsement under limited motivation or ability (Petty and Cacioppo, 1986).

In danmaku-enabled educational videos, these two routes are unlikely to operate only as independent predictors. Creator-related source cues, CV signals and peer feedback appear together in a dense and time-sensitive information environment. Users may rely on creator-related cues to reduce uncertainty about whether a video warrants attention. They may also use CV signals and peer feedback to assess whether deeper processing and real-time participation are worthwhile. Accordingly, this study advances an association-based sequential cross-route interpretation of ELM. Within the specified model sequence, creator-related SC cues are modeled as being linked with CV signals and user feedback interaction (UFI), which are then associated with RTE observed in danmaku traces. The COI framework (Garrison *et al.*, 2001) is used as a contextual lens for interpreting why engagement traces in learning-oriented environments often combine social and cognitive dimensions. It is not treated as a measurement model, but as an interpretive framework for understanding the nature of peer discourse.

2.3 *Research hypotheses*

This study operationalizes its constructs using publicly observable platform traces at the video level. Creator-related SC cues, CV signals, peer feedback interaction and RTE are derived from standard counters and text-derived indicators (comments and danmaku). Figures 1–3 provide interface context and an overview of key platform cues and conceptual routes; the hypotheses below specify the expected associations among constructs under a sequential cross-route logic rooted in ELM (Petty and Cacioppo, 1986).

2.3.1 Source credibility (SC). SC encompasses observable creator-level prominence signals, such as follower counts and historical performance metrics. In digital information environments, users often rely on these aggregated indicators as a heuristic proxy for authority and potential utility, rather than assessing credibility through direct expertise verification. Accordingly, we conceptualize SC as a peripheral cue that may reduce uncertainty and inform users' initial expectations and is therefore expected to be associated with subsequent engagement patterns (Li and See-To, 2024; Muda and Hamzah, 2021).

2.3.2 Content value (CV). Because this study utilizes archival platform traces, CV is captured through post-consumption appreciation metrics (e.g. likes, collections and coins). These indicators are treated as visible collective appreciation signals within the platform



Figure 1. Video playback interface

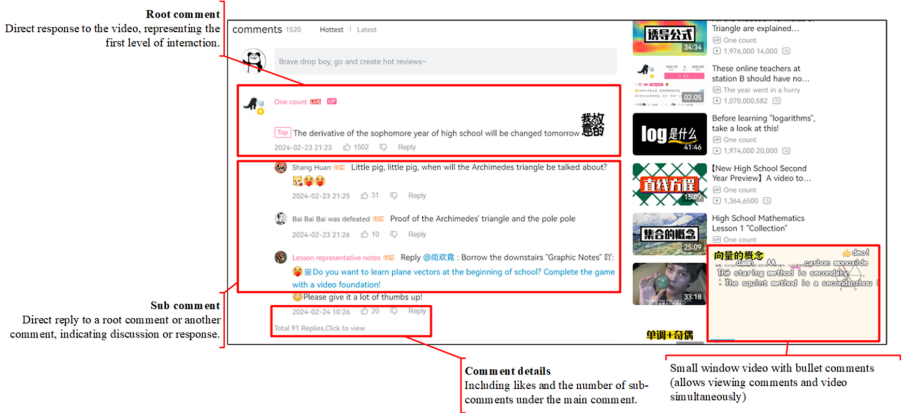


Figure 2. Video comment interface

environment. More broadly, perceived usefulness has been identified as a central rational driver of continued engagement in digitally mediated learning environments (Ke et al., 2026; Li et al., 2026). While these metrics are accumulated post-viewing, they serve as a visible quality signal within the information environment that users process in conjunction with source cues. Therefore, we interpret CV not as a temporal antecedent to the viewing act itself, but as a proxy for the intrinsic “informational currency” of the content that is visible within the user’s engagement environment.

H1 is stated accordingly:

H1. Source Credibility (SC) is positively associated with content value signals (CV).

2.3.3 User feedback interaction (UFI). UFI reflects users’ observable participation in the shared evaluation space around a video. In this study, UFI is captured via comment volume and related interaction signals (e.g. root comments, comment likes and sentiment-derived counts). These indicators represent the extent to which viewers externalize evaluations, request clarification, or respond to others, thereby making feedback an explicit part of the information environment (Phua et al., 2020).

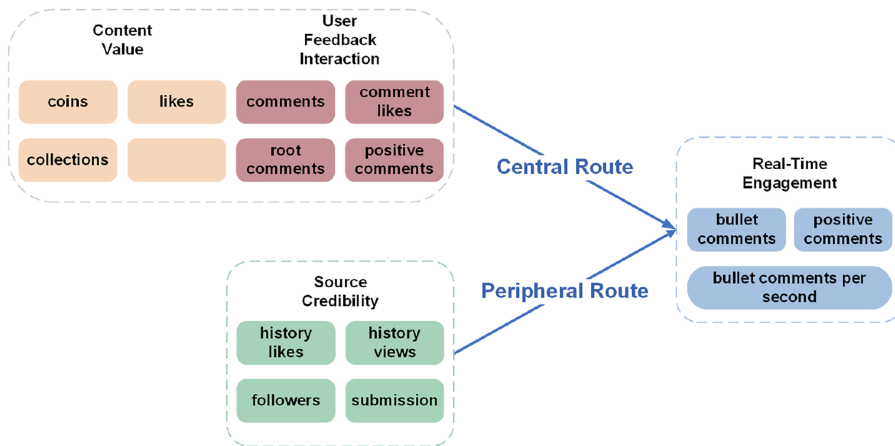


Figure 3. Conceptual factor and route diagram

Under the cross-route logic, SC may be associated with lower perceived risk of engaging and with entry into the feedback space, while stronger CV signals may be associated with user evaluations and questions (Oh *et al.*, 2018). Therefore, we state the following hypotheses in associational terms consistent with archival data:

H2. Source credibility (SC) is positively associated with user feedback interaction (UFI).

H3. Content value signals (CV) are positively associated with user feedback interaction (UFI).

2.3.4 Real-time engagement (RTE). RTE refers to users' during-view participation that is temporally anchored to the video stream. On danmaku-enabled platforms, RTE is reflected in the volume and density of danmaku as well as sentiment-derived counts, capturing the immediacy and intensity of real-time reactions (Zeng *et al.*, 2024). As a behavioral trace, RTE marks observable moments of active, time-sensitive expression while information is still unfolding.

Based on ELM, both peripheral cues, represented by SC and central cues, represented by CV signals and peer feedback interaction, can relate to real-time participation. Using archival data, we formulate the following hypotheses as associations rather than time-ordered causal claims:

H4. Source Credibility (SC) is positively associated with real-time engagement (RTE).

H5. Content value signals (CV) are positively associated with real-time engagement (RTE).

H6. User feedback interaction (UFI) is positively associated with real-time engagement (RTE).

2.3.5 Mediating hypothesis. To examine whether the observed associations are consistent with a *sequential cross-route* mechanism, we test a set of indirect-effect hypotheses among SC, CV, UFI and RTE. Conceptually, creator-related SC cues may be linked to initial expectations; this can be reflected in stronger CV signals and greater entry into the peer feedback space, which may in turn relate to more frequent real-time participation.

Because our indicators are archival behavioral traces aggregated at the video level, the indirect paths reported below should be interpreted as statistical decompositions of associations rather than causal mediation in the strict time-ordered sense. We therefore use

cautious language and treat the results as evidence of pattern consistency with the proposed sequential mechanism.

- H7. The associations among SC, CV, UFI and RTE exhibit indirect-effect patterns consistent with a sequential cross-route mechanism.
- H7a. SC is indirectly associated with RTE through CV and UFI ($SC \rightarrow CV \rightarrow UFI \rightarrow RTE$).
- H7b. SC is indirectly associated with RTE through CV ($SC \rightarrow CV \rightarrow RTE$).
- H7c. SC is indirectly associated with UFI through CV ($SC \rightarrow CV \rightarrow UFI$).
- H7d. SC is indirectly associated with RTE through UFI ($SC \rightarrow UFI \rightarrow RTE$).
- H7e. CV is indirectly associated with RTE through UFI ($CV \rightarrow UFI \rightarrow RTE$).

The hypothesized model of this study is illustrated in Figure 4.

2.3.6 *The moderating role of contextual factors.* While the ELM explains how users rely on central and peripheral cues, the relative salience of these cues is unlikely to remain constant across all video contexts. User-generated content platforms vary substantially in content performance, creator visibility and peer interaction dynamics. In danmaku-enabled educational videos, users encounter creator-related cues, CV signals, peer feedback and real-time commentary within the same information environment. These contextual conditions may alter the strength of the structural paths proposed in the engagement model. More broadly, evidence from digital process research indicates that the effects of digital technologies are not uniform but vary across functional stages and contextual conditions (Xu et al., 2025). We therefore examine three moderators that correspond to content context, source context and peer discourse context: Audience CE, Creator Standing (CS) and Danmaku Climate (DC).

Audience CE captures the extent to which a video context is accompanied by deliberate engagement. Conceptually, CE differs from CV. CV represents accumulated appreciation attached to a video, whereas CE reflects the contextual efficiency with which viewed content is associated with active responses. Under ELM, higher CE contexts are expected to show greater salience of substantive content cues because viewers are more likely to treat the video as worthy of continued attention and interaction. Lower CE contexts, in contrast, may show stronger alignment with SC and social heuristics in observable engagement patterns. Accordingly, we propose:

- H8. The strengths of the path relationships in the engagement model differ significantly between videos with high and low Audience Conversion Efficacy.

CS represents the broader platform position of the uploader. It extends beyond a single reputation cue by reflecting whether the creator has an established presence and historical

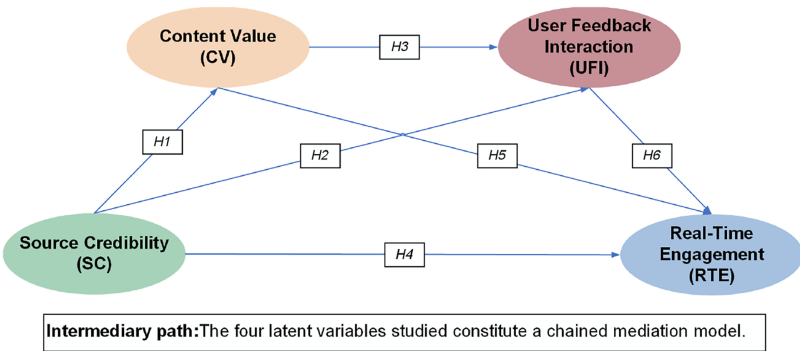


Figure 4. Hypothesized structural model

footprint in the platform community. As a source-context condition, high CS may correspond to differences in how viewers interpret SC and to lower uncertainty around entry into the feedback and danmaku spaces. Accordingly, we propose:

H9. The strengths of the path relationships in the engagement model differ significantly between videos from creators with high and low Creator Standing.

DC represents the affective tone of the real-time peer information environment. A constructive climate may be associated with peer feedback appearing more useful and socially safe, whereas a passive or mixed climate may weaken the informational value of the surrounding commentary (Wang and Chen, 2026). Because danmaku is processed alongside the focal video, this climate may condition how feedback and CV cues are associated with real-time participation. Accordingly, we propose:

H10. The strengths of the path relationships in the engagement model differ significantly between videos with a constructive and a passive or mixed Danmaku Climate.

For empirical testing, each moderator hypothesis is examined across the same six structural paths in the model. The suffixes a through f correspond respectively to SC → CV, SC → UFI, SC → RTE, CV → UFI, CV → RTE and UFI → RTE. Accordingly, the MGA results report H8a through H8f for CE, H9a through H9f for CS and H10a through H10f for DC.

3. Research methodology

3.1 Sample selection

Bilibili serves as a prominent ecosystem for examining HII, particularly within the dissemination of educational content. The platform is distinguished by its integration of rich learning resources with a synchronous “danmaku” comment mechanism, creating a dense social information layer (Wei *et al.*, 2023). This study targets the “Campus Learning” section under the “Knowledge” category, which hosts a spectrum of content ranging from formal instructional videos (e.g. exam analyses) to informal knowledge sharing (e.g. interview strategies). In this context, the interaction between creators and viewers extends beyond passive consumption and is associated with a collaborative learning dynamic linked to stronger engagement outcomes (Leem, 2023).

To ensure the sample represents high-utility information artifacts rather than merely popular entertainment, data collection was strictly prioritized based on “favorites” (collections) rather than view counts or likes. Within the framework of digital repositories, the “favorite” function operates as a personal knowledge management tool, explicitly signaling a user’s intent to archive, retrieve and reuse information (Wei *et al.*, 2023). In contrast to “likes”, which often capture fleeting affective responses, “favorites” indicate a deliberate cognitive judgment regarding the content’s long-term educational value and the user’s consolidation needs. Consequently, selecting samples via this archival metric helps focus the analysis on videos likely to have functioned as knowledge carriers, thereby supporting the validity of investigating deep information adoption behaviors.

3.2 Data collection

We curated a dataset of the 300 most-favorited videos within the “Campus Learning” section, identified by their unique BV numbers through the platform’s native filtering interface. The collection window was strategically limited to content published between December 10, 2023 and March 10, 2024. This specific three-month timeframe was selected to increase the likelihood that each video had accrued a stable viewership baseline post-publication, thereby reducing the transient inflation of view counts often observed in newly released content (Dong *et al.*, 2022), while simultaneously minimizing potential data biases arising from significant temporal discrepancies.

Following sample identification, comprehensive behavioral traces – including metadata for the 300 videos and their respective uploaders – were retrieved via Bilibili's open API. All collected data are publicly accessible, ensuring full compliance with ethical and legal standards for digital research. The final dataset presents a broad spectrum of engagement, with favorite counts ranging from 6,579 to 301,343, providing a robust empirical foundation for the subsequent structural analysis.

3.3 Data filtering and processing

Following initial data collection, a notable observation was made regarding videos that, while possessing a high number of favorites, lacked interaction. This suggests that some videos conveyed information but showed limited interactivity relative to the focus of this study. To align the sample with the study's focus on observable interaction, videos with fewer than 10 comments or danmaku were filtered out, resulting in a final selection of 261 videos for analysis.

The data processing for this study primarily involved sentiment analysis of the comments and danmaku texts. To ensure accurate text sentiment classification, a Chinese sentiment analysis model, RoBERTa-JD-Binary, fine-tuned on the JD review corpus, was utilized. This model has been reported to achieve high accuracy in Chinese sentiment classification tasks (Fei, 2024). The model was applied to perform binary classification of the sentiments (positive or negative) for 118,380 comments and 130,877 danmaku entries from the 261 video samples.

3.4 Construct operationalization and grouping variables

The constructs were operationalized using publicly observable video-level platform traces. SC was measured using uploader followers, historical likes, historical views and submissions. CV was measured using coins, collections and likes, which capture collective appreciation attached to the video. UFI was measured using comments, positive comments, root comments and total comment likes. RTE was measured using danmaku comments, positive danmaku comments and danmaku comments per second. These indicators were log-transformed before PLS-SEM estimation, as reported in the descriptive statistics section.

The three contextual grouping variables used in the MGA were also constructed at the video level. Audience CE was operationalized as the ratio of observable engagement actions to views and was calculated as follows: $CE = \frac{(Likes + Collections + Coins + Shares)}{Views}$. This grouping variable differs from CV because it normalizes engagement actions by exposure rather than representing the absolute level of accumulated appreciation. CS was operationalized as a composite standing index and was calculated as follows, where z denotes a standardized score: $CS = z(followers) + z(submissions) + z(historical\ views)$. DC was operationalized as the proportion of positive danmaku comments and was calculated as positive danmaku comments divided by total danmaku comments.

For the multi-group analysis, the sample was split into high and low groups using median-based cutoffs for each moderator. Because the sample included 261 videos, observations equal to or above the median were assigned to the high group, producing 131 videos in the high group and 130 videos in the low group for each comparison. Videos with $CE \geq 0.157$ were assigned to the high-CE group, and the remaining videos were assigned to the low-CE group. Videos with $CS \geq -0.837$ were assigned to the high CS group, and the remaining videos were assigned to the low CS group. Videos with $DC \geq 0.741$ were assigned to the constructive DC group, and the remaining videos were assigned to the passive or mixed DC group.

4. Results

4.1 Descriptive statistics

This study utilized archival data from Bilibili, drawing on publicly available video-level platform indicators and uploader information rather than survey responses. Archival data refers to data that has already been collected and preserved to support new research or analysis

(Fisher and Chaffee, 2018). Table 1 reports descriptive statistics for the original, untransformed platform indicators so that the empirical scale of the dataset remains transparent. The log-transformed indicators used for PLS-SEM estimation are reported separately in Appendix Table A1.

Prior to model estimation, the distributional properties of the indicators were assessed using the Shapiro–Wilk test. As shown in Table 1, the original platform indicators did not satisfy normality assumptions. Given this non-normality, PLS-SEM was selected as the primary analytical method. Unlike its covariance-based counterpart (CB-SEM), PLS-SEM is a variance-based, non-parametric approach that does not require normally distributed data, making it suitable for predictive modeling with non-normal behavioral trace data (Hair *et al.*, 2011).

For model estimation, natural logarithmic transformations were applied to reduce distributional skewness, alleviate scale differences and improve numerical stability. Where required for variables with possible zero values, a small pseudocount was used to ensure the feasibility of the logarithmic transformation, following the data-processing logic suggested by Biswas (2016). The transformed indicators were used for PLS-SEM estimation, whereas Table 1 preserves the original measurement scale.

4.2 Reflective measurement model assessment

This study employed PLS-SEM to evaluate real-time interactive behaviors in danmaku videos. This methodology is widely applied in user behavior research on online platforms, particularly excelling in predicting and explaining user behaviors (Klein *et al.*, 2015; Moghadamzadeh *et al.*, 2020). To validate the proposed model's effectiveness and reliability, an assessment of the measurement model was conducted.

The evaluation of the measurement model was based on three criteria: internal consistency, convergent validity and discriminant validity. As shown in Table 2, the Composite Reliability and Cronbach's Alpha coefficients for all latent variables exceeded 0.70, indicating good internal consistency (Hair *et al.*, 2021; Li *et al.*, 2024b; Xia *et al.*, 2024). Additionally, with the exception of the natural logarithm of submissions, all factor loadings surpassed 0.70 and the Average Variance Extracted (AVE) values exceeded 0.50, confirming the measurement model's convergent validity and indicating strong associations between observed variables and their corresponding latent variables (Fornell and Larcker, 1981; Wang *et al.*, 2024b, 2025).

For discriminant validity, the AVE square roots of each latent variable were compared against their correlation coefficients, following Fornell and Larcker's (1981) criteria (Fornell and Larcker, 1981; Wang *et al.*, 2024a). As shown in Table 3, all AVE square roots exceeded their respective correlation coefficients, demonstrating good discriminant validity. Furthermore, the Heterotrait-Monotrait Ratio (HTMT) and bootstrapping statistical tests further supported discriminant validity, with all HTMT values below the 0.85 threshold (Hair *et al.*, 2019). Confidence interval analyses indicated that none of the intervals included 1, further reinforcing the model's discriminant validity. These results provide a solid foundation for subsequent structural model analyses.

4.3 Structural model assessment

In the structural model assessment, SmartPLS 4 was utilized for analysis via PLS-SEM. Unlike CB-SEM, PLS-SEM is a predictive-oriented modeling approach that does not depend on model fit indices (Shmueli *et al.*, 2019). Therefore, fit indices such as SRMR and NFI require cautious application in PLS-SEM, and there is a lack of direct evidence supporting their use for model evaluation based on archival data. Consequently, model evaluation and selection should primarily rely on predictive indicators rather than traditional fit indices (Hair *et al.*, 2019, 2021). Accordingly, the assessment of the structural model focused primarily on predictive capability and the significance of path coefficients.

Table 1. Descriptive statistics for the original untransformed platform indicators

Item	Description	Min	Q1	Median	Q3	Max	Mean	SD	SW $p > 0.05$
followers	Followers of uploader	856	33,233	169,569	483,478	14,138,613	834,571.76	2,420,107.82	No
histLikes	Historical likes	2,460	172,854	596,663	1,482,732	33,941,536	2,293,136.71	5,739,761.82	No
histViews	Historical views	57,946	2,924,960	8,655,928	26,905,507	782,217,907	47,411,076.97	131,128,292.60	No
submissions	Submissions	1	59	141	343	2,135	257.90	340.12	No
coins	Coins	226	973	1,834	3,838	46,660	3,981.06	6,456.35	No
collections	Collections	6,579	8,720	11,957	23,405	301,343	23,023.68	36,241.29	No
likes	Likes	1,494	7,803	18,148	42,431	444,991	38,784.09	58,185.24	No
comments	Comments	14	115	268	485	5,382	453.56	657.14	No
positiveComments	Positive comments	9	89	200	383	3,867	348.92	490.94	No
rootComments	Root comments	9	81	177	350	4,436	349.14	555.47	No
totalCommentLikes	Total comment likes	19	789	2,340	8,137	114,001	8,754.23	16,262.40	No
bulletComments	Danmaku comments	10	49	144	428	7,200	501.44	1,022.80	No
positiveBulletComments	Positive danmaku comments	1	39	106	331	5,518	373.66	777.19	No
bulletCommentsPerSecond	Danmaku comments per second	<0.01	0.13	0.66	2.20	9.52	1.62	2.12	No

Note(s): $N = 261$. Q1/Q3 = quartiles; SD = standard deviation; SW = Shapiro–Wilk. Yes indicates $p > 0.05$; No indicates that normality was not supported

Table 2. Summary of the results for the measurement model

Latent variable	Item	Internal consistency		Convergent validity		Discriminant validity
		Composite reliability	Cronbach's alpha	Loadings	AVE	
		>0.70	>0.70	>0.70	>0.50	HTMT CI < 1 (upper <0.85)
Source credibility	The natural logarithm of the number of followers of the uploader	0.937	0.916	0.882	0.790	Yes
	The natural logarithm of historical likes			0.973		
	The natural logarithm of historical views			0.978		
Content value	The natural logarithm of submissions	0.822	0.716	0.691	0.608	Yes
	The natural logarithm of coins			0.785		
	The natural logarithm of collections			0.703		
User feedback interaction	The natural logarithm of likes	0.969	0.956	0.844	0.887	Yes
	The natural logarithm of comments			0.983		
	The natural logarithm of positive comments			0.968		
Real-time engagement	The natural logarithm of root comments	0.902	0.829	0.979	0.757	Yes
	The natural logarithm of total comment likes			0.828		
	The natural logarithm of bullet comments			0.943		
	The natural logarithm of positive bullet comments			0.937		
	The natural logarithm of bullet comments per second			0.710		

Table 3. Correlation matrix with AVE

	AVE	Source credibility	Content value	User feedback interaction	Real-time engagement
Source credibility	0.790	0.889	0.086	0.082	0.287
Content value	0.608	0.294	0.780	0.362	0.179
User feedback interaction	0.887	0.287	0.602	0.942	0.461
Real-time engagement	0.757	0.423	0.536	0.679	0.870

Note(s): 1) Lower triangle shows the correlation coefficients

2) Upper triangle shows the squared values of the correlation coefficients

3) Diagonal elements show the square roots of the average variance extracted (AVE)

4) The diagonal elements (square roots of AVE) should be greater than the corresponding inter-construct correlation coefficients

The structural model was first evaluated in terms of collinearity and explanatory power. Inner Variance Inflation Factor (VIF) values were used to assess potential multicollinearity among structural predictors, with values below 5 indicating no serious concern (Hair *et al.*, 2017). As shown in Table 4, all inner VIF values ranged from 1.000 to 1.610. The same table also reports R^2 and adjusted R^2 values for the endogenous constructs: CV ($R^2 = 0.087$, adjusted $R^2 = 0.083$), UFI ($R^2 = 0.376$, adjusted $R^2 = 0.371$) and RTE ($R^2 = 0.533$, adjusted $R^2 = 0.528$).

Because the structural model was estimated using standardized latent construct scores derived from log-transformed indicators, the path coefficients should be interpreted as standardized conditional associations among constructs. For example, the coefficient for CV $\rightarrow$ UFI indicates the expected standard-deviation change in UFI associated with a one-standard-deviation increase in CV, conditional on the other predictors in the model. These coefficients should not be read as raw-count changes in platform metrics or as direct elasticity estimates.

The bootstrapping procedure in SmartPLS 4 was then used to estimate the significance of the structural path coefficients. As reported in Table 5, all six hypothesized direct associations were statistically significant and the corresponding 95% confidence intervals did not include zero. Therefore, H1 through H6 are supported.

The structural model for this study is illustrated in Figure 5.

The bootstrapping procedure in SmartPLS 4 was also used to estimate the total, direct and indirect associations among SC, CV, UFI and RTE. Because the data are observational and aggregated at the video level, these results are interpreted as statistical decompositions of associations rather than as evidence of causal mediation.

As shown in Table 6, the total indirect association of SC with RTE was statistically significant ($\beta = 0.195$, $p < 0.001$, 95% CI [0.139, 0.260]), with VAF = 0.461, which is consistent with partial statistical mediation. At the specific-path level, H7a, H7c and H7e are supported. H7b and H7d show statistically significant indirect associations, but they are not interpreted as substantive mediation paths under the VAF threshold because their VAF values are <0.20 .

Table 4. Inner VIF and explanatory power of the structural model

	Content value $R^2 = 0.087$; adj. $R^2 = 0.083$	User feedback interaction $R^2 = 0.376$; adj. $R^2 = 0.371$	Real-time engagement $R^2 = 0.533$; adj. $R^2 = 0.528$
Source credibility	1.000	1.095	1.118
Content value	–	1.095	1.610
User feedback interaction	–	–	1.602

Note(s): Cells report inner VIF values; column headings report R^2 and adjusted R^2 for endogenous constructs

Table 5. Bootstrapped significance tests of standardized path coefficients

Hypothesis	Relationship	β	STDEV	t value	95% CI	p value	Decision
H1	SC $\rightarrow$ CV	0.294	0.049	5.988	[0.206, 0.400]	<0.001	Supported
H2	SC $\rightarrow$ UFI	0.120	0.044	2.732	[0.032, 0.202]	0.006	Supported
H3	CV $\rightarrow$ UFI	0.567	0.046	12.416	[0.478, 0.658]	<0.001	Supported
H4	SC $\rightarrow$ RTE	0.228	0.050	4.601	[0.128, 0.323]	<0.001	Supported
H5	CV $\rightarrow$ RTE	0.155	0.049	3.143	[0.058, 0.252]	0.002	Supported
H6	UFI $\rightarrow$ RTE	0.521	0.050	10.465	[0.416, 0.612]	<0.001	Supported

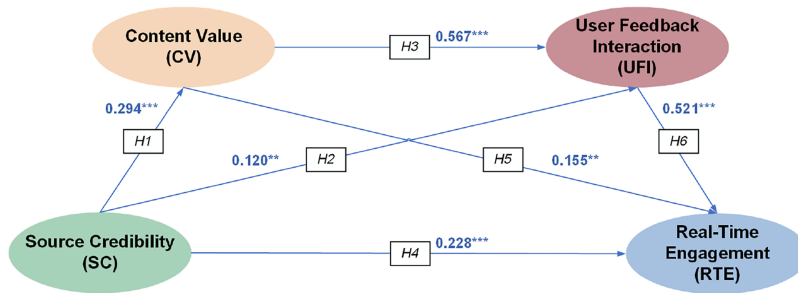


Figure 5. Structural model with standardized path coefficients. **Note.** Two asterisks indicate $p < 0.01$; three asterisks indicate $p < 0.001$

4.4 Multi-group analysis

To advance beyond a static model of engagement and to test H8, H9 and H10, a multi-group analysis (MGA) was conducted. The analysis examined whether the structural path relationships differed across subgroups defined by three contextual moderators: Audience CE, CS and DC. Before comparing path coefficients, the Measurement Invariance of Composite Models (MICOM) procedure was conducted to assess measurement equivalence across the subgroups (Henseler et al., 2016). As shown in Table 7, the three-step MICOM procedure indicated that at least partial measurement invariance was achieved across the CS and DC groupings. For the CE grouping, full or partial invariance was established for SC, CV and UFI. However, the RTE construct did not meet the compositional invariance criterion in the CE comparison. Therefore, CE-related path comparisons involving RTE should be interpreted with caution.

Following the measurement invariance assessment, MGA was conducted using the permutation test procedure in SmartPLS 4.1 to examine path-coefficient differences between groups. To align the results with the hypothesis numbering, each moderator was tested across the same six structural paths: a = SC → CV, b = SC → UFI, c = SC → RTE, d = CV → UFI, e = CV → RTE and f = UFI → RTE. Thus, Table 8 reports H8a through H8f for CE, H9a through H9f for CS and H10a through H10f for DC.

The results show that significant group differences emerged only for Audience CE. H8b was supported because the SC → UFI path was significant in the low-CE group ($\beta = 0.172$, $p < 0.05$) but not in the high-CE group ($\beta = -0.009$), with a significant difference between groups ($p = 0.037$). The H8c comparison also showed a significant between-group difference because the SC → RTE path was stronger in the low-CE group ($\beta = 0.316$, $p < 0.001$) than in the high-CE group ($\beta = 0.088$), with a significant difference between groups ($p = 0.017$). However, this RTE-related CE comparison should be interpreted cautiously in light of the MICOM result reported above. H8d was supported because the CV → UFI path was stronger in the high-CE group ($\beta = 0.684$, $p < 0.001$) than in the low-CE group ($\beta = 0.490$, $p < 0.001$), with a significant difference between groups ($p = 0.030$). No other path coefficient differed significantly across the CE, CS, or DC groupings.

4.5 Thematic analysis of danmaku content

While the multi-group analysis shows that some model path relationships differ across contextual factors, it does not explain the underlying discursive patterns associated with these variations. To provide a complementary thematic account, we conducted BERTopic-based topic modeling on the danmaku corpus. The danmaku texts were aggregated and cleaned before topic modeling. Sentence embeddings were generated using the multilingual SentenceTransformer model paraphrase-multilingual-MiniLM-L12-v2. Chinese tokenization was implemented through a customized Jieba tokenizer within CountVectorizer, together with predefined Chinese stop-word filtering. BERTopic was configured with multilingual language

Table 6. Bootstrapped indirect associations and mediation assessment

Hypothesis/ association	Relationship	Estimate	STDEV	t value	95% CI	p value	VAF	Interpretation
Total association	SC → RTE	0.423	0.049	8.714	[0.329, 0.521]	<0.001	N/A	N/A
Total association	SC → UFI	0.287	0.045	6.436	[0.204, 0.376]	<0.001	N/A	N/A
Total association	CV → RTE	0.450	0.045	9.986	[0.362, 0.538]	<0.001	N/A	N/A
Direct association	SC → RTE	0.228	0.050	4.601	[0.128, 0.323]	<0.001	N/A	N/A
Overall indirect association H7a H7b H7c H7d H7e	SC → RTE	0.195	0.031	6.317	[0.139, 0.260]	<0.001	0.461	Partial statistical mediation
	SC → CV → UFI → RTE	0.087	0.017	4.997	[0.057, 0.126]	<0.001	0.206	Supported; partial statistical mediation
	SC → CV → RTE	0.046	0.018	2.592	[0.016, 0.086]	0.010	0.109	Significant indirect association, but not substantive mediation because VAF <0.20
	SC → CV → UFI	0.167	0.032	5.290	[0.113, 0.238]	<0.001	0.582	Supported; partial statistical mediation
	SC → UFI → RTE	0.062	0.024	2.607	[0.017, 0.109]	0.009	0.147	Significant indirect association, but not substantive mediation because VAF <0.20
	CV → UFI → RTE	0.295	0.035	8.470	[0.229, 0.368]	<0.001	0.656	Supported; partial statistical mediation
Note(s): STDEV = bootstrap standard deviation; VAF = variance accounted for; N/A = not applicable								

Table 7. Measurement invariance testing results using MICOM

Grouping variable	Construct	Step 1	Step 2	Comp. inv. (<i>p</i> -value)	Step 3(a)		Step 3(b)		Final invariance status
		Configurational invariance	Correlation c-value		Difference (mean)	95% CI (mean)	Difference (variance)	95% CI (variance)	
Conversion efficacy	CV	Established	0.976	0.065	−0.041	−0.248, 0.236	−0.166	−0.388, 0.375	Full Invariance
	RTE	Established	0.992	0.013	0.403	−0.248, 0.242	0.193	−0.295, 0.273	Invariance Not Established
	SC	Established	0.991	0.121	0.454	−0.242, 0.245	0.202	−0.335, 0.362	Partial Invariance
Creator standing	UFI	Established	1.000	0.894	0.232	−0.244, 0.231	0.070	−0.342, 0.340	Full Invariance
	CV	Established	0.990	0.264	−0.144	−0.232, 0.242	0.080	−0.394, 0.392	Full Invariance
	RTE	Established	0.996	0.102	−0.387	−0.239, 0.228	−0.382	−0.297, 0.289	Partial Invariance
	SC	Established	0.998	0.732	−1.393	−0.250, 0.239	0.080	−0.360, 0.337	Partial Invariance
Danmaku climate	UFI	Established	1.000	0.079	−0.261	−0.248, 0.251	0.123	−0.324, 0.335	Partial Invariance
	CV	Established	0.976	0.074	0.002	−0.255, 0.245	−0.145	−0.356, 0.365	Full Invariance
	RTE	Established	1.000	0.804	−0.103	−0.241, 0.241	−0.119	−0.294, 0.284	Full Invariance
	SC	Established	0.999	0.730	0.048	−0.228, 0.234	−0.165	−0.368, 0.371	Full Invariance
	UFI	Established	1.000	0.712	0.076	−0.233, 0.252	0.104	−0.343, 0.353	Full Invariance

Note(s): The *p*-value for Step 2 must be > 0.05 to establish compositional invariance. This criterion is decisive even in cases where the 95% confidence interval may contain zero

Table 8. Comprehensive results of multi-group analysis (MGA) on path coefficients

Hypothesis	Path	Grouping variable	Low group coefficient	Low group 95% CI	High group coefficient	High group 95% CI	Coefficient differences	p-value difference	Difference supported
H8a	SC → CV	CE	0.372***	0.214, 0.532	0.300**	0.188, 0.448	−0.072	0.456	No
H9a		CS	0.417***	0.323, 0.533	0.373***	0.259, 0.498	−0.044	0.652	No
H10a		DC	0.339***	0.243, 0.477	0.261**	0.090, 0.423	−0.078	0.420	No
H8b	SC → UFI	CE	0.172*	0.026, 0.299	−0.009	−0.137, 0.119	−0.181	0.037	Yes
H9b		CS	0.007	−0.146, 0.148	0.150*	0.025, 0.268	0.143	0.110	No
H10b		DC	0.099	−0.032, 0.211	0.128*	0.000, 0.257	0.029	0.741	No
H8c	SC → RTE	CE	0.316***	0.171, 0.456	0.088	−0.054, 0.214	−0.228	0.017	Yes
H9c		CS	0.210***	0.082, 0.337	0.232***	0.098, 0.367	0.022	0.825	No
H10c		DC	0.148*	0.002, 0.278	0.292***	0.162, 0.424	0.144	0.147	No
H8d	CV → UFI	CE	0.490***	0.344, 0.654	0.684***	0.596, 0.763	0.194	0.030	Yes
H9d		CS	0.513***	0.353, 0.673	0.617***	0.525, 0.707	0.104	0.266	No
H10d		DC	0.587***	0.496, 0.701	0.584***	0.450, 0.708	−0.003	0.981	No
H8e	CV → RTE	CE	0.121†	−0.017, 0.253	0.272**	0.087, 0.436	0.151	0.126	No
H9e		CS	0.111	−0.036, 0.281	0.157*	0.012, 0.296	0.046	0.685	No
H10e		DC	0.248***	0.127, 0.368	0.095	−0.057, 0.250	−0.153	0.120	No
H8f	UFI → RTE	CE	0.476***	0.327, 0.614	0.497***	0.347, 0.640	0.021	0.826	No
H9f		CS	0.562***	0.400, 0.693	0.501***	0.365, 0.628	−0.061	0.583	No
H10f		DC	0.552***	0.424, 0.671	0.479***	0.308, 0.630	−0.073	0.479	No

Note(s): CE, Audience Conversion Efficacy; CS, Creator Standing; DC, Danmaku Climate

Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$. Italic p -values indicate a significant difference between groups

processing, the customized vectorizer, a minimum topic size of 50 and verbose output enabled. No custom UMAP or HDBSCAN settings were specified in the archived workflow; therefore, the dimensionality-reduction and density-based clustering stages followed BERTopic's default pipeline. These settings make the topic modeling procedure traceable while avoiding the introduction of unsupported hyperparameters.

The exported topic summary contained 34 rows, including the noise topic (–1) and 33 non-noise topics, with topic IDs, topic sizes, model-generated topic labels, representative keywords and representative documents. Intertopic distance and topic word-score visualizations were also generated to support topic inspection. To assess topic interpretability, overall topic coherence was calculated as $C_v = 0.364$ using a C_v -style indirect confirmation procedure across the eligible topic set. Given the short, colloquial and highly context-dependent nature of danmaku comments, this coherence value was treated as a diagnostic indicator of topic interpretability rather than as standalone validation (Wang *et al.*, 2024b, 2025). The model-generated topics were subsequently inspected and aggregated through inductive coding into five overarching theme categories: cognitive and content, metacognitive and navigational, affective and motivational, social and community and off-topic and photic.

The thematic distribution across the moderating conditions is presented in Figure 6. As depicted in Figure 6(a), the analysis shows pronounced thematic differences across CE groups. In the high-CE group, danmaku discourse was characterized by Cognitive and Content themes (26.7%), a proportion more than double that of the low-CE group (12.5%). Conversely, Social and Community themes were more prevalent in the low-CE group (40.0%) than in the high-CE group (26.7%). A similar pattern was observed for Affective & Motivational content, which was more common in low-efficacy contexts (20.0% vs. 16.7%). This thematic divergence provides an association-based interpretation of the MGA findings. The stronger presence of substantive content in high-efficacy videos is consistent with the stronger CV → UFI association (H8d) in this group; when danmaku discourse is more oriented toward the material itself, the perceived value of that content appears more closely linked with further interaction. In contrast, the prevalence of social and affective commentary in low-efficacy videos is consistent with the stronger associations involving the peripheral cue of SC with both UFI (H8b) and RTE (H8c). When the content itself shows less content-oriented engagement, discourse places greater emphasis on the social environment and heuristic cues like creator reputation as part of the observed interaction pattern. While thematic differences were also observed across CS and DC groups (Figure 6b and 6c), they did not correspond to statistically significant path differences in the MGA, supporting CE as the most evident contextual moderator in this dataset.

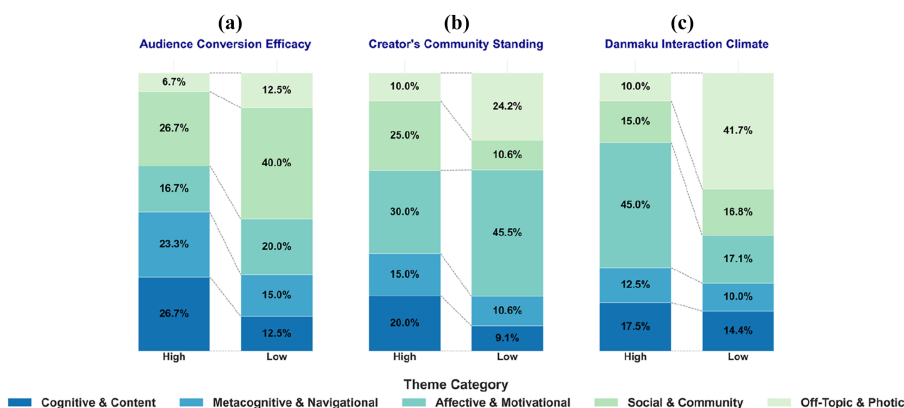


Figure 6. Thematic distribution of danmaku content across contextual moderators

5. Discussion

5.1 Modeling observed engagement patterns: from peripheral heuristics to central engagement

This study employed the ELM to examine user engagement patterns in information-rich, user-generated content (UGC) environments. The central design tension in such systems is a core Human-Computer Interaction (HCI) challenge: socio-technical features like *danmaku* commentary, intended to enhance social presence, risk inducing cognitive overload and a split-attention effect by presenting a constant stream of peripheral information (Peng *et al.*, 2025; Zhang *et al.*, 2025a).

The ELM's dual-route framework provides a useful lens for interpreting observable engagement in environments saturated with competing signals. The findings suggest that SC should not be treated merely as a parallel variable, but as an orientation cue or usability scaffold (Wang, 2015). In high-density information streams, SC may serve as a heuristic filter through which viewers assess whether an information source warrants attention. When higher SC is accompanied by stronger CV, content quality can be interpreted as a value signal associated with active processing (Shoufan, 2019), while peer feedback (UFI) is associated with the social validation surrounding engagement (Lee *et al.*, 2024). This pattern is consistent with a sequential cross-route account ($SC \rightarrow CV \rightarrow UFI \rightarrow RTE$): creator reputation is linked with CV, which is then associated with social interaction and RTE. This refines prior ELM research by suggesting that, in goal-oriented information seeking, peripheral cues may need to be accompanied by perceived content efficacy to be associated with meaningful behavioral participation.

Beyond estimating the direct associations of both peripheral and central cues, the findings indicate a cross-route pattern that can inform the design of user journeys in complex interactive systems. This pattern suggests a refinement of the ELM in this context (Beautemps and Bresges, 2021) and offers a granular, association-based account for linking the COI framework with HCI principles. The analysis suggests that SC, a peripheral cue, does not function merely as a simple heuristic; instead, it is associated with an initial usability-scaffold role that may help viewers orient themselves before engaging in more effortful central processing. In a UGC learning context, a credible creator may be perceived as a trustworthy instructor (Zha *et al.*, 2018) and this perception can be associated with a psychologically safe and cohesive group climate – a socio-emotional foundation that may make deeper cognitive engagement more accessible (Maré and Mutezo, 2025). This association pattern is further clarified by the chain mediation analysis, which provides an empirical basis for adaptive interface design. The full mediation path, $SC \rightarrow CV \rightarrow UFI \rightarrow RTE$, was statistically significant (H7a), representing an observed association pattern from peripheral UI cues to CV and feedback interaction. This pathway can be interpreted as an observable learning-oriented sequence: source-related interface cues are associated with CV, which is then linked with sustained UFI and observable RTE (Garrison *et al.*, 2001).

Crucially, the weaker substantive support for the simpler indirect paths, namely $SC \rightarrow CV \rightarrow RTE$ and $SC \rightarrow UFI \rightarrow RTE$, helps clarify how the present findings both align with and differ from prior work. Existing studies on *danmaku* and online video interaction have shown that real-time comments, viewer cues and social presence can shape engagement and narrative involvement (Zhang *et al.*, 2025a, b). Studies of SC also suggest that credible sources can be associated with stronger engagement (Li and See-To, 2024; Yu *et al.*, 2026), while ELM research has long recognized that central and peripheral routes may influence one another (Lee, 2012). The present results extend this literature by suggesting that, in *danmaku*-enabled educational videos, SC is not best interpreted as an isolated engagement cue. Its association with RTE appears to be embedded in a broader sequence in which source cues are linked with CV and UFI. Social influence therefore operates less as a simple volume effect and more as a context-sensitive information layer through which users evaluate whether participation is worth sustaining. This interpretation remains associational because the archival data cannot establish the temporal order of users' internal processing.

5.2 An evidence-based principle for adaptive interface design

The study's primary contribution to HCI and information management is the identification of Audience CE not merely as a performance metric, but as a moderator of observed information-processing-related engagement patterns. As visualized in [Figure 7](#), the associations among source cues, CV, feedback interaction and RTE vary across content-efficacy contexts, offering an empirical basis for context-aware interface adaptation ([Keehr and Reardon, 2025](#)).

Specifically, the CE comparison indicates a context-contingent association pattern rather than evidence of a universal difference in processing routes. In lower-CE contexts, the stronger SC $\rightarrow$ UFI and SC $\rightarrow$ RTE associations suggest that SC and social signals may function as uncertainty-reduction cues when content efficacy is less evident. This pattern is consistent with the role of the social layer as an orientation aid in dense information streams, where affective and community cues can help users decide whether further participation is worthwhile ([Du and Huang, 2025](#)). However, it should not be interpreted as direct evidence that users abandon content processing.

Conversely, in higher-CE contexts, the stronger CV $\rightarrow$ UFI association suggests that CV is more closely linked with feedback interaction. This result is consistent with a content-oriented interpretation of engagement, where peer exchange appears more connected with semantic discussion than with social connectedness alone ([Zhu et al., 2025](#)). It does not establish that users directly invest cognitive effort or that content alone accounts for engagement. Rather, high-CE conditions appear to make CV more salient in the observed relationship between content evaluation and feedback interaction.

This structural divergence is further supported by the thematic analysis of the danmaku metadata. In high-efficacy videos, the stronger presence of Cognitive and Content themes is consistent with the stronger CV $\rightarrow$ UFI association, suggesting that peer exchange may support content-oriented engagement. In low-efficacy videos, the greater presence of Social and Community and Affective themes is consistent with the stronger role of source and social cues as heuristic anchors. This pattern cautions against treating denser danmaku or higher interaction volume as inherently better ([Zhao et al., 2026](#)). In educational settings, engagement metrics should be interpreted alongside whether interaction supports comprehension, retention, knowledge organization, or self-regulated learning, because learning analytics research shows that digital trace indicators are most meaningful when related to learners' goals, cognitive engagement and learning outcomes ([Xu et al., 2026](#)). From an attention economy perspective, interface optimization should avoid turning learning spaces into systems that maximize attention capture at the expense of learner welfare ([De La Torre et al., 2026](#)). More broadly, adaptive designs should be evaluated for whether they distribute learning support equitably rather than reproducing visibility bias or unequal access to useful guidance ([Albuquerque et al., 2026](#)).

Accordingly, this study proposes a principle of Dynamic Information Presentation. For lower-CE content, interfaces may foreground peripheral metadata, such as creator credentials, historical performance indicators and social proof elements, to support orientation and discovery. For higher-CE content, interfaces may reduce peripheral distractions and prioritize features that support semantic analysis and knowledge organization, such as threaded discussions or content-anchored annotations, so that interaction is more likely to support learning-oriented engagement. This design logic is consistent with recent human-centered adaptive learning research, which suggests that adaptive systems can adjust learning strategies based on behavioral indicators and learner-related psychological states while still requiring controlled validation before causal effectiveness claims can be made ([Lin et al., 2025](#)).

Finally, the non-significant moderation by CS and DC defines the boundary of this design principle. The results do not imply that creator position or DC are irrelevant. Rather, within this dataset, CE is the contextual condition most clearly associated with differences in the structural paths. Prior work suggests that creator type, influencer context ([Dong et al., 2025](#)) and real-time information flow can shape digital engagement in ways that may not be fully captured by aggregate standing or sentiment-ratio indicators ([Wu, 2025](#); [Yuan et al., 2025](#)).

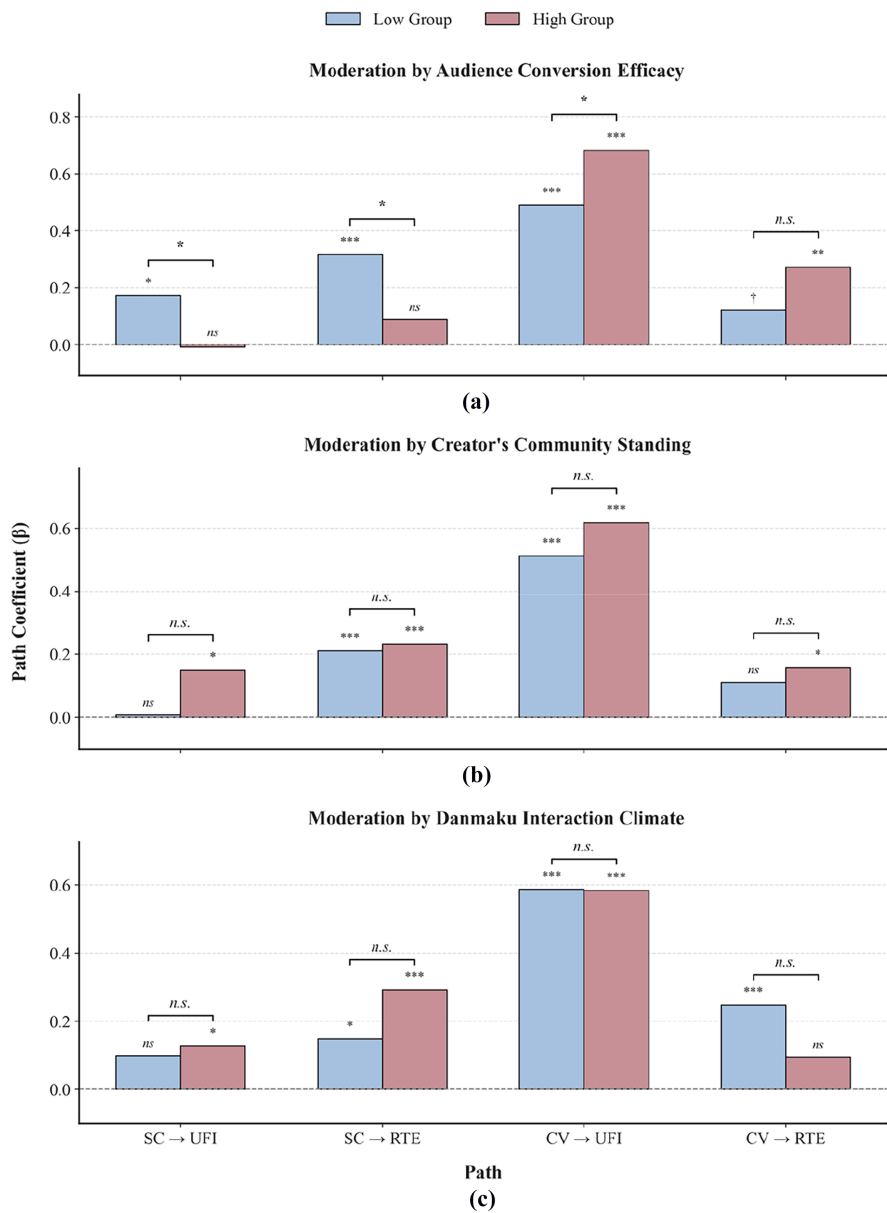


Figure 7. Multi-group analysis results for the three contextual moderators

Future research should therefore use richer measures of creator reputation and peer climate to examine whether these social-context variables condition engagement beyond the operationalization used in this study.

6. Conclusion and limitations

This study examined danmaku engagement as a form of HII in educational videos. Using behavioral trace data from Bilibili, it modeled how SC, CV, UFI and RTE are associated within a sequential cross-route pattern. The findings suggest that engagement in danmaku-enabled educational videos is not only a matter of isolated source, content, or peer cues. Rather, observable participation is more coherently explained as a linked process in which creator-related cues, content evaluation, feedback interaction and real-time expression operate together within an information-dense viewing environment.

The study also clarifies the role of Audience CE as a contextual condition for this process. In lower-CE contexts, the stronger associations involving SC suggest that source and social cues may play a greater role when the informational value of the content is less immediately evident. In higher-CE contexts, the stronger CV → UFI association suggests that peer interaction is more closely connected with content-oriented engagement. This distinction refines the interpretation of social influence in danmaku environments. Social cues are not merely associated with greater engagement volume; their role appears to depend on whether the surrounding content context supports heuristic orientation or content-focused exchange.

These findings have implications for platform and interface design, but they should be interpreted cautiously. Adaptive educational-video systems should not optimize only for visible engagement indicators such as danmaku volume, comments, clicks, or other activity counts. Such indicators may reflect meaningful knowledge exchange, but they may also reflect distraction, social following, or attention fragmentation. A more appropriate design principle is to align interface affordances with informational efficacy. Source and peer cues may be useful when learners need orientation and uncertainty reduction, whereas high-efficacy educational content may benefit from reduced peripheral noise and stronger support for semantic discussion, annotation and knowledge organization.

For educational practice, the results point to a distinction between behavioral engagement and learning-oriented engagement. This study did not directly measure learning outcomes, so it cannot claim that the observed engagement pathways improve comprehension, retention, transfer, or self-regulated learning. However, the model can inform future evaluations of educational-video interfaces by identifying when engagement is more likely to reflect content-oriented interaction rather than activity volume alone. From an equity perspective, adaptive systems should also avoid privileging only already visible creators, highly active users, or high-performing videos. They should support learners with different cognitive styles, attention capacities, prior knowledge, emotional responses and resource needs by providing manageable information environments.

Several limitations define the boundaries of these conclusions. First, the study uses archival behavioral trace data at the video level, so the indicators are indirect proxies for internal cognitive states rather than direct measures of attention, comprehension, or learning. Second, the cross-sectional and observational design limits causal and temporal inference. The reported paths should therefore be interpreted as statistical associations consistent with the proposed cross-route logic, not as evidence of a causal processing sequence. Third, the empirical setting is limited to Bilibili educational videos, and the generalizability of the model should be examined across other platforms, cultural contexts and instructional formats. Future research should combine trace data with learning assessments, experiments, longitudinal designs, surveys, eye-tracking, cognitive load measures, or other process-level evidence and should evaluate whether adaptive interface designs improve learning quality rather than only increasing observable engagement.

Data availability statement

All data used in this study were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Table A1. Descriptive statistics for the log-transformed PLS-SEM indicators

Item	Description	Min	Q1	Median	Q3	Max	Mean	SD	SW <i>p</i> > 0.05
Ln_followers	Log followers of uploader	6.75	10.41	12.04	13.09	16.46	11.76	2.04	No
Ln_histLikes	Log historical likes	7.81	12.06	13.30	14.21	17.34	13.12	1.88	No
Ln_histViews	Log historical views	10.97	14.89	15.97	17.11	20.48	16.04	1.82	Yes
Ln_submissions	Log submissions	0.00	4.08	4.95	5.84	7.67	4.84	1.34	No
Ln_coins	Log coins	5.42	6.88	7.51	8.25	10.75	7.62	1.07	No
Ln_collections	Log collections	8.79	9.07	9.39	10.06	12.62	9.62	0.76	No
Ln_likes	Log likes	7.31	8.96	9.81	10.66	13.01	9.85	1.18	No
Ln_comments	Log comments	2.64	4.74	5.59	6.18	8.59	5.50	1.11	Yes
Ln_positiveComments	Log positive comments	2.20	4.49	5.30	5.95	8.26	5.24	1.13	Yes
Ln_rootComments	Log root comments	2.20	4.39	5.18	5.86	8.40	5.18	1.15	Yes
Ln_totalCommentLikes	Log total comment likes	2.94	6.67	7.76	9.00	11.64	7.76	1.79	Yes
Ln_BulletComments	Log danmaku comments	2.30	3.89	4.97	6.06	8.88	5.09	1.48	No
Ln_positiveBulletComments	Log positive danmaku comments	0.69	3.69	4.67	5.81	8.62	4.78	1.52	Yes
Ln_bulletCommentsPerSecond	Log danmaku comments per second	−8.36	−2.06	−0.42	0.79	2.25	−0.87	2.29	No

Note(s): *N* = 261. These indicators were used for PLS-SEM estimation; original-scale statistics are reported in [Table 1](#)

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